Linear Functions & Graphing — Answer Key

Type 1 Slope from Two Points

1. $\dfrac{7-(3)}{4-(2)} = \dfrac{4}{2}$

Answer: $m = 2$

2. $\dfrac{1-(5)}{3-(1)} = \dfrac{-4}{2}$

Answer: $m = -2$

3. $\dfrac{6-(4)}{2-(-2)} = \dfrac{2}{4}$

Answer: $m = \dfrac{1}{2}$

4. $\dfrac{-3-(-3)}{5-(0)} = \dfrac{0}{5}$ (no rise: horizontal line)

Answer: $m = 0$

5. $\dfrac{8-(-1)}{2-(-4)} = \dfrac{9}{6}$

Answer: $m = \dfrac{3}{2}$

6. $\dfrac{5}{0}$: you cannot divide by 0. A vertical line.

Answer: undefined (vertical line)

Type 2 Slope from a Graph

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7. From $(-2, -3)$ to $(2, 5)$: rise 8, run 4.

Answer: $m = 2$

8. From $(-3, 4)$ to $(3, 0)$: rise -4, run 6.

Answer: $m = -\dfrac{2}{3}$

9. From $(0, 1)$ to $(4, 1)$: rise 0, run 4. Flat line.

Answer: $m = 0$

10. From $(-4, -2)$ to $(4, 4)$: rise 6, run 8.

Answer: $m = \dfrac{3}{4}$

Type 3 Intercepts

11. $y = 0$: $x = 6$. $x = 0$: $y = 4$.

Answer: $(6, 0)$ **and** $(0, 4)$

12. $y = 0$: $x = 4$. $x = 0$: $y = -3$.

Answer: $(4, 0)$ **and** $(0, -3)$

13. $y = 0$: $x = 2$. $x = 0$: $y = 5$.

Answer: $(2, 0)$ **and** $(0, 5)$

14. $y = 0$: $x = -6$. $x = 0$: $y = 4$.

Answer: $(-6, 0)$ **and** $(0, 4)$

15. $y = 0$: $x = -8$. $x = 0$: $y = -4$.

Answer: $(-8, 0)$ **and** $(0, -4)$

16. $y = 0$: $x = 5$. $x = 0$: $y = 3$.

Answer: $(5, 0)$ **and** $(0, 3)$

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Type 4 Slope-Intercept Form

17. Already in $y = mx + b$ form. $y = 3x - 4$

Answer: $m = 3,\ b = -4$

18. Already in $y = mx + b$ form. $y = -\frac{1}{2}x + 5$

Answer: $m = -\frac{1}{2},\ b = 5$

19. Subtract $2x$. $y = -2x + 7$

Answer: $m = -2,\ b = 7$

20. Subtract $3x$, divide by 3. $y = -x + 3$

Answer: $m = -1,\ b = 3$

21. Subtract $4x$, divide by -2. $y = 2x - 4$

Answer: $m = 2,\ b = -4$

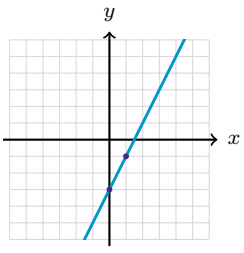
22. Subtract x, divide by -3. $y = \frac{1}{3}x - 2$

Answer: $m = \frac{1}{3},\ b = -2$

Type 5 Graph $y = mx + b$

23. Plot $(0, -3)$, then use the slope 2: up 2, right 1.

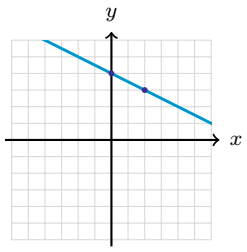
Answer: $y = 2x - 3$



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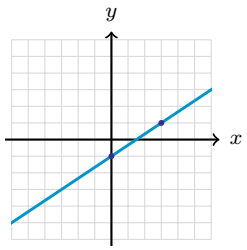
24. Plot $(0, 4)$, then use the slope $-\frac{1}{2}$: down 1, right 2.

Answer: $y = -\frac{1}{2}x + 4$



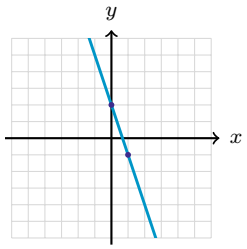
25. Plot $(0, -1)$, then use the slope $\frac{2}{3}$: up 2, right 3.

Answer: $y = \frac{2}{3}x - 1$



26. Plot $(0, 2)$, then use the slope -3: down 3, right 1.

Answer: $y = -3x + 2$



Type 6 Linear Functions in Real Life

27. $C = 2x + 3;\ 2 \cdot 12 + 3 = 27$

Answer: $C = 2x + 3$**, $27**

28. $h = 30 - 2t;\ 0 = 30 - 2t$ gives $t = 15$

Answer: $h = -2t + 30$**, 15 hours**

29. $h = 1.5w + 4;\ 1.5 \cdot 6 + 4 = 13$

Answer: $h = 1.5w + 4$**, 13 in**

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30. Start \$200 (b); loses \$15 each week ($m = -15$).

Answer: $M = -15w + 200$**; slope:** $-\$15$ **per week; intercept: \$200 at the start**

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